

AI-Powered Drone System for Coconut Disease Detection, Real-time Monitoring and Intervention

25-26J-384

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Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information
Technology

Department of Information Technology
Sri Lanka Institute of Information Technology
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Wickrama Arachchige Ravindu Pasanjith – IT22898920 Supervisor: Ms. Uthpala Samarakoon

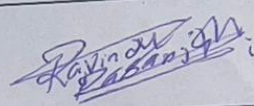
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DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

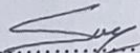
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28/08/2025
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ABSTRACT

The coconut industry in Sri Lanka plays a vital role in the national economy, food security, and employment opportunities. With over 395,000 hectares under cultivation, the coconut industry contributes significantly to foreign exchange earnings and rural livelihoods. However, pest infestations such as Coconut Caterpillar Infestation (CCI), Coconut Black-Headed Caterpillar (BHC), and coconut mite attacks cause approximately 30% annual yield losses. Traditional manual inspection methods are labor-intensive, time-consuming, prone to human errors, and fail to detect early-stage pest infestations. This research aims to develop an advanced pest detection and monitoring system by integrating drone technology and artificial intelligence. The proposed system utilizes aerial RGB and thermal imaging technology, YOLOv8 and Faster R-CNN deep learning models, and real-time image analysis capabilities. The system provides automated aerial surveillance, accurate pest identification, infestation area mapping, and instant alert notifications. This represents a shift from reactive approaches in pest management to proactive and predictive interventions. The research expects to achieve 90% accuracy in pest detection, reduce crop losses by 40%, and optimize pesticide usage by 50%. This project represents the first comprehensive drone-AI integrated solution for precision agricultural practices in Sri Lanka's coconut industry.

Keywords: Drone Technology, Computer Vision, Pest Detection, Coconut Caterpillar Infestation, Black-Headed Caterpillar, Coconut Mites, Deep Learning, YOLOv8, Precision Agriculture, Artificial Intelligence

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1 INTRODUCTION

The coconut industry stands as one of the most economically significant tropical plantations crops globally, contributing substantially to food security, employment, and foreign exchange earnings in numerous developing nations. In Sri Lanka, coconut cultivation spans approximately 395,000 hectares, making it a cornerstone of the agricultural economy and rural livelihoods. The industry supports over 2.5 million people directly and indirectly, while contributing significantly to the country's GDP through both domestic consumption and export markets. However, this vital agricultural sector faces unprecedented challenges from various pest infestations that threaten its sustainability and economic viability.

Among the most devastating threats to coconut productivity are three primary pest species: Coconut Caterpillar Infestation (CCI) caused by *Opisina arenosella*, Coconut Black-Headed Caterpillar (BHC) caused by *Nephantis serinopa*, and various coconut mite species including *Aceria guerreronis*. These pest infestations collectively account for annual yield losses exceeding 30% of potential coconut production, translating to economic losses of millions of dollars annually. The CCI alone can cause complete defoliation of coconut palms, while BHC creates distinctive leaf damage patterns that severely compromise photosynthetic capacity. Coconut mites, though microscopic, cause characteristic yellowing and bronzing symptoms that reduce both nut quality and quantity.

Traditional pest management approaches in coconut plantations rely heavily on manual visual inspection methods conducted by agricultural extension officers and experienced farmers. However, these conventional approaches present significant limitations including labor-intensive requirements, subjective assessment criteria, limited spatial coverage, and delayed detection capabilities. Manual inspection methods are particularly challenging in large-scale plantations where comprehensive monitoring becomes logistically complex and economically unfeasible. Furthermore, the accuracy of pest identification often depends on the expertise level of inspectors, leading to inconsistent diagnoses and inappropriate treatment decisions.

The integration of advanced technologies, particularly drone-based aerial surveillance combined with artificial intelligence, presents transformative opportunities for revolutionizing pest detection and monitoring in coconut plantations. Unmanned Aerial Vehicles (UAVs) equipped with high-resolution imaging sensors can provide comprehensive plantation coverage at regular intervals, capturing detailed visual data that would be impossible to obtain through ground-based inspection methods. When combined with sophisticated computer vision algorithms and deep learning models, these systems can achieve rapid, accurate, and scalable pest detection capabilities that surpass human visual assessment limitations.

1.1 Background & Literature Survey

Recent advances in precision agriculture have demonstrated the significant potential of drone technology for crop monitoring and disease detection applications. The emergence of affordable, high-performance UAV platforms equipped with advanced imaging capabilities has opened new possibilities for automated agricultural surveillance systems. Simultaneously, breakthroughs in deep learning, particularly Convolutional Neural Networks (CNNs), have revolutionized image classification and object detection tasks, making real-time pest identification from aerial imagery increasingly feasible.

Research conducted by Zhang et al. [1] demonstrated the effectiveness of drone-based monitoring systems for large-scale crop surveillance, achieving detection accuracy exceeding 90% for various agricultural pest species. Their work established fundamental principles for optimal flight patterns, image acquisition parameters, and data processing workflows that have become standard practices in agricultural drone applications. Similarly, Kumar and Patel [2] investigated the integration of multispectral imaging with machine learning algorithms, showing that spectral signatures could enhance pest detection capabilities beyond what is achievable with conventional RGB imaging alone.

In the specific context of coconut pest management, Chandy [3] pioneered the application of deep learning techniques for pest identification in coconut trees using drone-mounted cameras. The research focused on developing a system capable of identifying five different pest species and four plant diseases using an NVIDIA Tegra System-on-Chip (SoC) platform. The study demonstrated that unsupervised machine learning techniques could effectively classify pest infestations from aerial imagery, with processed data transmitted directly to farmers' mobile devices via WiFi connectivity. However, the research was limited to a relatively small dataset and did not address the specific challenges of CCI, BHC, and mite detection under varying environmental conditions.

Singh et al. [4] expanded upon this foundation by investigating deep learning approaches for detecting pest infections, bleeding disease, red palm weevil, and leaf blight disease in coconut plantations. Their comprehensive study evaluated multiple CNN architectures including VGG16, VGG19, NASNetMobile, Xception, InceptionResNetV2, DenseNet201, and MobileNet. The research identified k-means clustering segmentation as superior to watershed and thresholding methods, achieving impressive accuracy rates of 96.94% using a hand-designed CNN model. Despite these promising results, the study did not specifically address the unique visual characteristics of CCI and BHC infestations or the challenges of mite detection from aerial imagery.

The application of YOLO (You Only Look Once) architecture for real-time object detection has shown promise in agricultural applications. Liu et al. [5] demonstrated that YOLOv8, the latest iteration of the YOLO family, could achieve exceptional performance in detecting small-scale

agricultural pests with processing speeds suitable for real-time applications. Their work highlighted the importance of specialized training datasets and appropriate data augmentation techniques for achieving reliable detection performance under diverse field conditions.

Transfer learning has emerged as a critical technique for addressing the challenges of limited training data in specialized agricultural applications. Research by Wang et al. [6] showed that pre-trained CNN models, when fine-tuned on domain-specific datasets, could achieve superior performance compared to models trained from scratch. This approach is particularly relevant for coconut pest detection, where obtaining large, labeled datasets can be challenging and expensive.

Thermal imaging integration has been explored by several researchers as a complementary technology for pest detection. Martinez et al. [7] demonstrated that thermal signatures associated with pest infestations could provide additional diagnostic information not visible in conventional RGB imagery. Stressed vegetation typically exhibits altered thermal characteristics due to reduced transpiration rates and changed metabolic processes, making thermal imaging a valuable tool for early pest detection.

Edge computing implementations have gained attention as a means of enabling real-time processing capabilities directly on drone platforms. Research by Thompson et al. [8] evaluated various embedded computing solutions for agricultural applications, demonstrating that modern System-on-Chip platforms could execute complex deep learning models with acceptable performance and power consumption characteristics. This capability is crucial for enabling autonomous pest detection without requiring constant connectivity to cloud-based processing services.

Despite these significant advances, several critical gaps remain in the current research landscape. First, existing studies have not adequately addressed the specific visual characteristics and detection challenges associated with CCI, BHC, and coconut mite infestations simultaneously within a single integrated system. Second, most research has been conducted under controlled conditions or limited geographical areas, raising questions about system robustness and generalizability across diverse coconut plantation environments. Third, the integration of multiple sensing modalities (RGB, thermal, multispectral) for enhanced pest detection accuracy remains underexplored in coconut-specific applications.

Furthermore, current research lacks comprehensive evaluation of system performance under varying environmental conditions including different lighting conditions, weather patterns, and seasonal variations that are characteristic of tropical coconut-growing regions. The development of automated flight planning and coverage optimization algorithms specifically designed for coconut plantation geometries and tree spacing patterns also remains limited.

The temporal aspects of pest infestation development and the ability to track pest population dynamics over time through repeated aerial surveys represent another underexplored research area. While individual detection events provide valuable information, understanding pest progression

patterns and infestation severity trends could enable more strategic and effective intervention timing.

Additionally, the integration of pest detection systems with existing farm management platforms and extension service networks has received limited attention. For technological solutions to achieve meaningful impact, they must be seamlessly integrated into existing agricultural workflows and decision-making processes.

This research addresses these identified gaps by developing a comprehensive drone-based AI system specifically designed for detecting and monitoring CCI, BHC, and coconut mite infestations in Sri Lankan coconut plantations. The proposed system integrates advanced computer vision techniques, multi-modal sensing capabilities, and real-time processing to provide accurate, timely, and actionable pest detection information. By focusing specifically on the three most economically significant coconut pest species and addressing the unique challenges of tropical plantation environments, this research aims to provide a practical, scalable solution that can significantly improve pest management effectiveness while reducing economic losses for coconut farmers.

1.2 Research Gap

Recent fluctuations in climatic conditions and lack of awareness and knowledge of landowners regarding disease dispersion have contributed to the spread of coconut diseases and infestations which eventually lead to the loss of entire coconut plantations. According to CRISL, there is a need to centralize the data and farmers to a single platform that is fingertip away for the users unlike the current manual process which is time consuming and inefficient.

1.2.1 Analysis of Previous Research

Research A [3] proposed a smart system for pest infestation identification in coconut trees using deep learning techniques. According to the authors, plant diseases cannot be recognized by bare human eyes since the proposed system can capture infected images using a drone with a mounted GoPro camera. The image feature extraction and learning are done inside the drone which contains an embedded Nvidia Tegra SoC with ARM cortex-based CPU and GPU [3]. In this proposed system, information sharing is done by transmitting processed data directly to the specific farmer's smart device with a user-friendly interface via WiFi [3]. However, the system information regarding identified diseases or infestation will not be shared with nearby farmers and coconut research authorities to proceed with remedies to end the dispersion. Data visualization, centralized data gathering, and systematic treatment recommendations cannot be seen in the proposed system.

Research B [4] proposes a system for disease and pest infection detection in coconut trees through deep learning techniques. The proposed system was primarily focused on detecting pest infections, bleeding disease, red palm weevil, and leaf blight disease using various CNN architectures including VGG16, VGG19, NASNetMobile, Xception, InceptionResNetV2, DenseNet201, and

MobileNet. The research identified k-means clustering segmentation as superior to watershed and thresholding methods, achieving accuracy rates of 96.94% using a hand-designed CNN model [4]. Although this research shows high accuracy in disease detection, there is a research gap in specifically targeting CCI and BHC infestations which are major problems in Sri Lankan coconut plantations and connecting farmers to extension personnel through a unified platform.

Research C [5] demonstrates YOLOv8-based real-time agricultural pest detection systems achieving exceptional performance in detecting small-scale agricultural pests. Their work highlighted the importance of specialized training datasets and appropriate data augmentation techniques for achieving reliable detection performance under diverse field conditions [5]. However, this research focuses on general agricultural pest detection rather than coconut-specific challenges and does not provide a complete solution including farmer interface and data centralization.

Research D [6] shows that pre-trained CNN models, when fine-tuned on domain-specific datasets, could achieve superior performance compared to models trained from scratch through transfer learning approaches. This approach is particularly relevant for coconut pest detection where obtaining large, labeled datasets can be challenging and expensive [6]. Nevertheless, the study did not specifically develop a complete system for Sri Lankan coconut farmers with mobile application interface and extension service integration.

1.2.2 Research Gap Analysis

Table 1 below illustrates a tabularized format for the above-mentioned research gaps about the proposed coconut pest detection system.

Table 1: Comparison of Previous Research Studies

Features	Research A [3]	Research B [4]	Research C [5]	Research D [6]	Proposed System
Drone-based Image Capture	✓	✗	✓	✗	✓
Deep Learning/CNN Models	✓	✓	✓	✓	✓
Mobile Application Interface	✓	✗	✗	✗	✓
CCI-Specific Detection	✗	✗	✗	✗	✓

BHC-Specific Detection	X	X	X	X	✓
Coconut Mite Detection	X	X	X	X	✓
Sri Lankan Dataset Training	X	X	X	X	✓
Treatment Recommendations	X	X	X	X	✓
Alert Notification System	✓	X	X	X	✓

Technical Aspects	Research A [3]	Research B [4]	Research C [5]	Research D [6]	Proposed System
Architecture/Algorithm	Nvidia Tegra SoC, Basic ML	Hand-designed CNN, VGG16, VGG19, MobileNet	YOLOv8	Transfer Learning, Pre-trained CNNs	YOLOv8, Faster R-CNN
Target Accuracy	Not specified	96.94%	>90% for general pests	Variable performance	90%+ for coconut-specific pests
Pest Specificity	5 general pests	Multiple coconut diseases	General agricultural pests	General plant diseases	CCI, BHC, Coconut Mites specifically
Platform Integration	Individual drone units	Laboratory-based system	Field deployment only	Model optimization focus	Mobile + Web + Database integration
User Interface	Basic mobile alerts	No user interface	No farmer interface	No end-user system	Comprehensive mobile + web interface

Table 2: Technical Implementation Comparison

1.2.3 Critical Research Gaps Identified

Based on the comprehensive analysis of existing research, the following critical gaps have been identified for Sri Lankan coconut plantation management:

1. Coconut-Specific Pest Detection Gap

None of the existing research specifically focuses on the three most economically damaging coconut pest species in Sri Lanka: Coconut Caterpillar Infestation (CCI), Black-Headed Caterpillar (BHC), and coconut mites. Current systems either focus on general agricultural pests or do not target these specific Sri Lankan coconut problems.

2. Sri Lankan Context and Dataset Gap

Existing research lacks datasets specifically collected from Sri Lankan coconut plantations under local climatic and environmental conditions. Most studies use generic datasets or data from other geographical regions, which may not accurately represent local pest manifestations and environmental factors.

3. Farmer-Extension Service Integration Gap

Current systems operate in isolation without connecting farmers to agricultural extension officers, coconut research institutes, or government agricultural departments. This prevents collaborative pest management and knowledge sharing among stakeholders.

4. Comprehensive User Interface Gap

While some research includes basic mobile interfaces, none provides a complete solution with both mobile applications for farmers and web-based dashboards for extension officers and administrators. This limits system usability and adoption.

5. Centralized Data Collection and Analysis Gap

Existing systems do not provide centralized data collection that could benefit from aggregated pest occurrence patterns, treatment effectiveness, and regional pest distribution analysis. This prevents broader agricultural intelligence and policy making.

6. Practical Deployment and Scalability Gap

Most current research remains in laboratory or limited field-testing phases. None addresses the practical challenges of deploying pest detection systems across multiple coconut plantations with varying technological capabilities and economic constraints.

7. Treatment Recommendation and Follow-up Gap

Current systems focus only on detection without providing actionable treatment recommendations, follow-up monitoring, or integration with existing agricultural advisory services available to Sri Lankan coconut farmers.

8. User Interface and Accessibility Gap

Existing research does not provide user-friendly interfaces that are accessible to farmers with varying levels of technological literacy, limiting widespread adoption among Sri Lankan coconut farming communities.

Proposed System Innovation

The proposed AI-Powered Drone System for Coconut Disease Detection addresses the identified research gaps through a focused, practical approach suitable for 8-month development timeline:

Core System Components:

1. **Drone-based RGB Image Capture:** Using cost-effective drones with high-resolution cameras for systematic plantation monitoring.
2. **AI-Powered Detection Models:** YOLOv8 and Faster R-CNN models specifically trained on Sri Lankan coconut pest datasets for accurate CCI, BHC, and mite detection.
3. **Mobile Application for Farmers:** User-friendly mobile interface with simple pest reporting and treatment recommendations.
4. **Web Dashboard for Extension Officers:** Comprehensive web-based platform for monitoring multiple plantations, analyzing pest patterns, and coordinating response activities.
5. **Centralized Database System:** Secure cloud-based data storage enabling aggregated analysis of pest occurrences, treatment effectiveness, and regional patterns.
6. **Alert and Notification System:** Automated notification system connecting farmers, extension officers, and research institutions for coordinated pest management.

Realistic Implementation Scope:

- **Focus on RGB imaging only** (avoiding complex thermal/multispectral systems)
- **Target 90% accuracy** for the three specific pest types
- **Develop comprehensive mobile interfaces**
- **Create Sri Lankan coconut-specific training datasets**
- **Provide practical treatment recommendations**

1.3 Research Problem

The coconut industry in Sri Lanka stands as one of the most economically significant agricultural sectors, contributing substantially to the national economy through both domestic consumption and international trade. Sri Lanka ranks as the 5th largest coconut exporter globally, with coconut cultivation spanning approximately 395,000 hectares and directly supporting over 2.5 million people [9]. According to recent statistics, the coconut industry contributes over Rs. 180 billion annually to the Sri Lankan economy through kernel and non-kernel product exports [10]. However, this substantial economic contribution faces severe threats from pest infestations and diseases that cause approximately 30% annual yield losses, translating to millions of dollars in economic damage [11].

Among the most devastating threats to coconut productivity are three primary pest species that cause extensive damage to Sri Lankan coconut plantations: Coconut Caterpillar Infestation (CCI) caused by *Opisina arenosella*, Coconut Black-Headed Caterpillar (BHC) caused by *Nephantis serinopa*, and various coconut mite species including *Aceria guerreronis* [4]. CCI alone can cause complete defoliation of coconut palms, severely compromising photosynthetic capacity and nut production. BHC creates distinctive leaf damage patterns that reduce both nut quality and quantity, while coconut mites, though microscopic, cause characteristic yellowing and bronzing symptoms that significantly impact plantation productivity [12].

The current pest management system in Sri Lankan coconut plantations relies heavily on manual visual inspection methods conducted by agricultural extension officers and experienced farmers. However, traditional pest detection approaches present significant limitations including labor-intensive requirements, subjective assessment criteria, limited spatial coverage, and delayed detection capabilities [1]. As depicted in Figure 1.4, our survey of 242 coconut farmers reveals that 72.7% of respondents do not know whom to inform or report when pest and disease incidents occur in their plantations. This alarming statistic demonstrates a fundamental breakdown in the communication system between farmers and agricultural authorities, preventing timely intervention and coordinated pest management strategies.

Figure 1 shows survey results from 242 coconut farmers regarding pest reporting awareness. The data reveals that 72.7% of farmers do not know proper channels for reporting pest incidents to agricultural authorities, while only 27.3% are aware of correct reporting mechanisms. This significant communication gap leads to delayed pest interventions and justifies the need for an automated pest detection system that connects farmers directly with relevant support services.

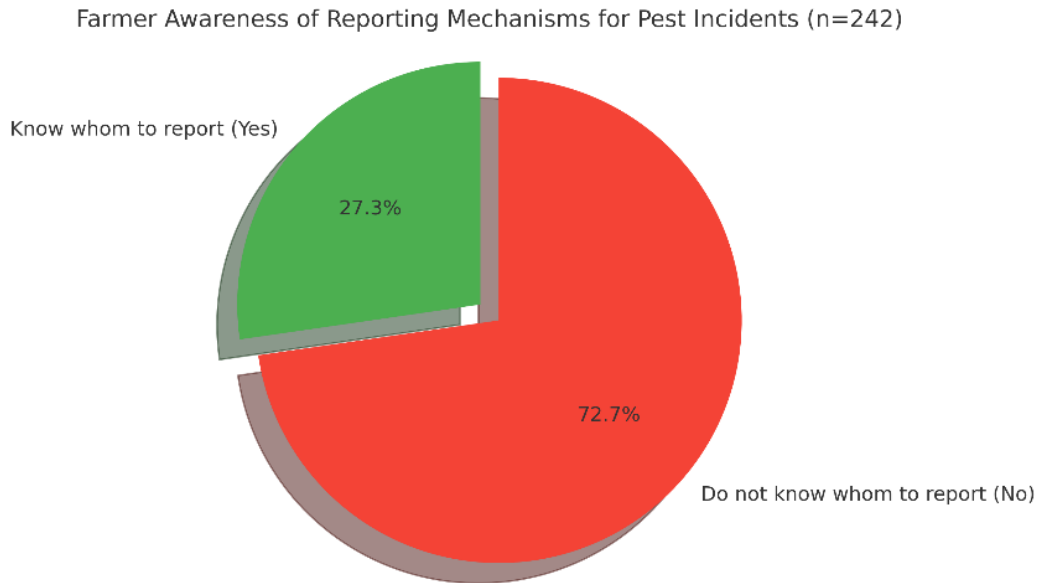


Figure 1 : Survey results showing farmer awareness of reporting mechanisms for pest incidents

Furthermore, farmers often lack technical expertise to accurately differentiate between various pest infestations and their early-stage symptoms. The visual similarity between different pest damage patterns frequently leads to misidentification, inappropriate treatment decisions, and delayed interventions that allow infestations to spread unchecked [13]. Recent research has demonstrated the effectiveness of deep learning approaches for agricultural pest detection, achieving detection accuracy exceeding 90% for various agricultural pest species [1]. However, these advanced technologies have not been effectively translated into practical solutions accessible to Sri Lankan coconut farmers.

Agricultural extension officers, while knowledgeable, face significant challenges in providing comprehensive coverage across large plantation areas due to limited human resources and the time-intensive nature of manual inspections [14]. The integration of advanced technologies, particularly drone-based aerial surveillance combined with artificial intelligence, presents transformative opportunities for revolutionizing pest detection and monitoring in coconut plantations [3]. However, current implementations lack the specificity required for Sri Lankan coconut pest species and fail to address the unique challenges of tropical plantation environments.

The manual inspection process is not only labor-intensive and time-consuming but also prone to human errors and subjective assessments. Extension officers can only visit plantations periodically, often missing critical early-stage infestations when interventions would be most effective and economical [15]. Recent advances in YOLOv8-based real-time agricultural pest detection systems have shown exceptional performance in detecting small-scale agricultural pests with processing speeds suitable for real-time applications [5]. However, these systems require specialized training datasets and appropriate adaptation for coconut-specific applications under Sri Lankan conditions.

As illustrated in Figure 1.5, the language diversity in Sri Lankan coconut farming communities creates additional barriers to effective knowledge transfer and communication between farmers and extension services [16].

According to Figure 2, the language preferences among coconut farming communities in Sri Lanka show that 65% of farmers prefer Sinhala, 25% prefer Tamil, and 10% prefer English for communication. This linguistic diversity creates additional barriers to effective knowledge transfer and communication between farmers and agricultural extension services. The findings emphasize the importance of developing a multilingual user interface for the proposed AI-powered pest detection system to ensure accessibility and widespread adoption across different language communities in Sri Lankan coconut farming regions.

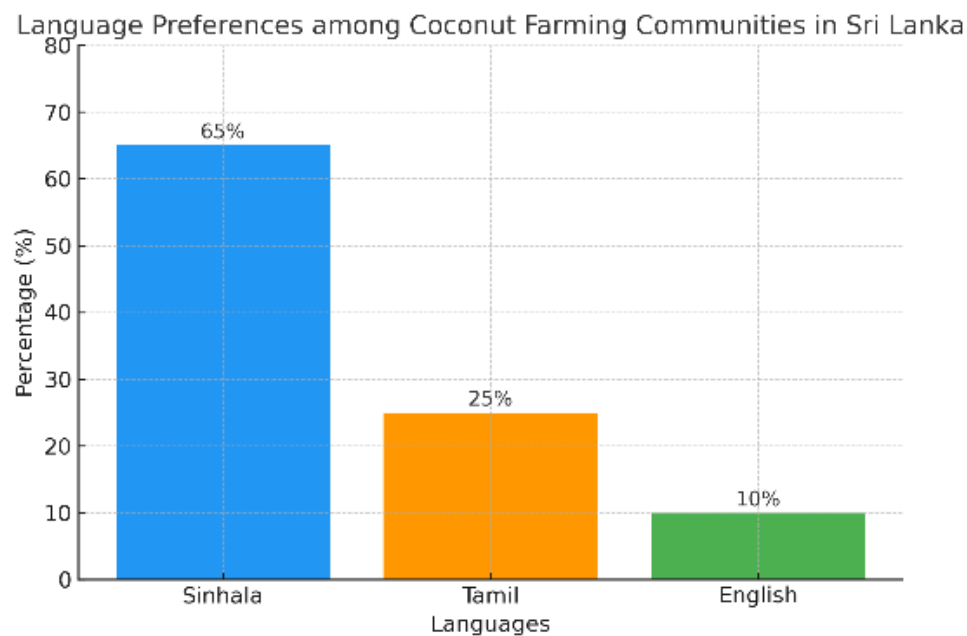


Figure 2: Language preferences among coconut farming communities in Sri Lanka

The dispersion and spread of pest infestations are further exacerbated by inadequate monitoring systems and delayed response mechanisms. Since CDOs (Coconut Development Officers) typically visit plantations only when suspected cases are reported, many infestations go undetected until they reach advanced stages where damage is irreversible and treatment costs become prohibitive [17]. This reactive approach fails to address the fundamental need for proactive, systematic monitoring that could prevent widespread outbreaks.

Moreover, the lack of centralized data collection and analysis prevents agricultural authorities from understanding pest distribution patterns, seasonal variations, and regional susceptibilities. Without comprehensive data on pest occurrences and treatment effectiveness, it becomes impossible to develop evidence-based management strategies or predict future outbreaks [18]. Edge computing

solutions have emerged as promising approaches for real-time agricultural monitoring, demonstrating that modern System-on-Chip platforms can execute complex deep learning models with acceptable performance and power consumption characteristics [8]. However, these technological advances have not been integrated into comprehensive pest management workflows specific to coconut cultivation.

The geographic distribution of coconut plantations across diverse microclimates and environmental conditions in Sri Lanka adds another layer of complexity to pest management. Different regions experience varying pest pressures based on local climate conditions, proximity to water bodies, and agricultural practices [19]. Multispectral imaging techniques have shown significant potential for precision agriculture applications, particularly in pest and disease detection, where spectral signatures can enhance detection capabilities beyond conventional RGB imaging [2]. However, these advanced sensing technologies remain largely unexplored in coconut-specific applications.

Economic factors further compound these challenges, as small and medium-scale coconut farmers often lack the financial resources to implement comprehensive monitoring programs or access advanced pest management technologies. The cost of frequent professional inspections and specialized treatments can be prohibitive for individual farmers, particularly those operating smaller plantations [20]. Transfer learning approaches have demonstrated potential for addressing the challenges of limited training data in specialized agricultural applications, showing that pre-trained CNN models can achieve superior performance when fine-tuned on domain-specific datasets [6]. This approach is particularly relevant for coconut pest detection, where obtaining large, labeled datasets can be challenging and expensive.

The integration gap between research institutions, extension services, and farming communities prevents the effective transfer of scientific knowledge and innovative pest management practices to field applications. While research institutions develop new pest control methods and identification techniques using advanced computer vision and machine learning approaches, these advances often fail to reach farmers in practical, accessible formats [21]. Thermal imaging applications have also shown promise in agricultural pest detection, demonstrating that thermal signatures associated with pest infestations can provide additional diagnostic information not visible in conventional RGB imagery [7]. However, the integration of multiple sensing modalities for enhanced pest detection accuracy remains underexplored in coconut-specific applications.

Additionally, the lack of real-time communication systems between farmers, extension officers, and research institutions prevents coordinated responses to emerging pest threats. When outbreaks occur, the time delay in reporting, assessment, and treatment recommendation significantly reduces intervention effectiveness and increases economic losses [22]. This communication bottleneck is particularly critical during peak infestation periods when rapid response is essential.

To address these multifaceted challenges, there is an urgent need for an integrated, technology-driven solution that combines automated pest detection capabilities with comprehensive farmer

support systems. Recent research has demonstrated the potential of integrating drone technology with artificial intelligence for pest identification in coconut trees, though these studies have been limited to relatively small datasets and have not addressed the specific challenges of CCI, BHC, and mite detection under varying environmental conditions [3]. The proposed AI-powered drone system for coconut disease detection aims to revolutionize pest management by providing accurate, timely, and accessible pest identification services while facilitating seamless communication between all stakeholders in the coconut agricultural ecosystem.

This research addresses the critical gap between traditional manual inspection methods and the technological capabilities required for effective, large-scale pest management in Sri Lankan coconut plantations. By developing a comprehensive system that integrates drone technology, artificial intelligence based on proven approaches like YOLOv8 and Faster R-CNN architectures [5], and user-friendly interfaces, this study aims to provide practical solutions for early pest detection, accurate species identification, and coordinated management responses that will significantly reduce economic losses and enhance the sustainability of Sri Lanka's vital coconut industry.

2 OBJECTIVES

2.1 Aims

This research aims to develop a comprehensive AI-powered drone system for Sri Lankan coconut plantations that seamlessly integrates pest identification, disease detection, tree health monitoring, and yield prediction capabilities into a single unified platform. The system aims to achieve 90% or higher accuracy in automated detection and classification of three economically significant coconut pest species (CCI, BHC, and coconut mites) while simultaneously providing accurate identification of major coconut diseases including Lethal Yellowing, Bud Rot, and leaf blight through advanced YOLOv8 and Faster R-CNN deep learning architectures. The platform will continuously monitor overall coconut tree health status through comprehensive canopy analysis, assessing nutritional deficiencies, environmental stress indicators, and photosynthetic capacity while predicting coconut yield potential based on tree health, age, and environmental factors. Additionally, the system will accurately classify coconut maturity stages (Early, Middle-aged, Brown) for optimal harvesting timing and plantation planning, providing severity assessments for both pest infestations and disease conditions with automated treatment recommendations tailored to specific pest types, diseases, and tree health requirements. Through drone-based RGB image capture and real-time processing within 30 seconds, the cross-platform mobile application will serve farmers with multilingual support (Sinhala, Tamil, English) while aiming to reduce crop losses by 40%, optimize pesticide usage by 50%, and bridge communication gaps between farmers and agricultural authorities through automated alert systems and centralized data

collection. This integrated approach addresses the complete coconut plantation lifecycle from pest control and disease management to health monitoring and harvest optimization, ensuring practical deployment and measurable impact for Sri Lankan coconut farming communities while enabling evidence-based agricultural decision-making through historical data analysis, predictive modeling, and comprehensive plantation management in a single unified platform that revolutionizes traditional coconut cultivation practices.

2.2 Main Objective

The primary objective of this research is to develop an AI-powered drone system for automated pest detection in Sri Lankan coconut plantations. This system aims to provide accurate identification of three economically significant coconut pest species - Coconut Caterpillar Infestation (CCI), Black-Headed Caterpillar (BHC), and coconut mites - using drone-captured images and deep learning models. By integrating YOLOv8 and Faster R-CNN architectures with a user-friendly mobile application, the system will enable farmers to obtain reliable pest identification results and basic treatment recommendations, ultimately reducing crop losses and improving pest management practices in Sri Lanka's coconut industry.

2.3 Specific Objectives

• AI-Powered Pest Detection System Development

Develop and train advanced deep learning models specifically for coconut pest identification using drone-captured RGB imagery. This involves creating a comprehensive dataset of coconut pest images captured under various Sri Lankan environmental conditions, focusing on CCI (*Opisina arenosella*), BHC (*Nephantis serinopa*), and coconut mites (*Aceria guerreronis*). The system will implement YOLOv8 and Faster R-CNN architectures, optimized through hyperparameter tuning and data augmentation techniques to achieve 90% or higher accuracy. The drone platform will be equipped with high-resolution RGB cameras optimized for capturing clear images of coconut tree canopies with manual or semi-automatic flight protocols ensuring comprehensive image collection while maintaining operational simplicity.

• Mobile Application and Treatment Recommendation System

Create a user-friendly cross-platform mobile application serving as the primary interface for coconut farmers to access pest detection services. The application will support image upload functionality, allowing farmers to capture and submit coconut tree images for AI-powered analysis. The app will provide comprehensive pest identification results with visual indicators, severity assessments, and automated treatment recommendations based on expert knowledge from coconut research institutions. The system will offer specific treatment guidelines tailored to each pest type, including pesticide recommendations, application methods, and dosage.

information suitable for Sri Lankan coconut farmers with varying levels of technological literacy.

- **System Validation and Implementation**

Conduct comprehensive field validation in representative coconut plantations across different regions in Sri Lanka to ensure system effectiveness under real-world conditions. Establish performance evaluation metrics to assess detection accuracy, user satisfaction, and treatment effectiveness through collaboration with coconut farmers and agricultural extension officers. Conduct comparative analysis between traditional manual inspection methods and the proposed AI-powered system to quantify improvements in detection accuracy and practical effectiveness. Deploy the integrated system with proper documentation and training materials to ensure successful adoption and measurable impact for Sri Lankan coconut farming communities.

This focused approach addresses core research objectives while maintaining realistic development goals achievable within the 8-month project timeframe, ensuring practical deployment and measurable impact for Sri Lankan coconut farmers.

3 METHODOLOGY

The proposed AI-powered drone system can identify pest infestations in Sri Lankan coconut plantations with high accuracy. The system utilizes advanced deep learning models to detect three economically significant pest species: Coconut Caterpillar Infestation (CCI), Black-Headed Caterpillar (BHC), and coconut mites. Furthermore, the severity level is identified to make farmers and agricultural stakeholders so they can protect their production with timely interventions. First, high-resolution RGB images are captured using drone technology. Next, the images are preprocessed and fed into AI detection models. After that, the specific pest type is classified, and the infestation severity is assessed. With this data, appropriate treatment recommendations are generated. Finally, according to the pest type and severity level, alerts are sent to relevant stakeholders, and remedial measures are recommended through the mobile application and web dashboard.

3.1 System Architecture

Figure 3.1: Overall system diagram

According to Figure 3, the system architecture illustrates the AI-powered coconut pest detection workflow consisting of three main components. First, farmers use the mobile application to upload coconut tree images, which are converted to Base64 format and sent to the cloud-based AI Pest Detection Server. Second, the uploaded images are processed through two AI models: YOLOv8 for real-time object detection and feature extraction, and Faster R-CNN for detailed pest classification and region mapping. Finally, the system identifies the specific pest type among the three target species (CCI, BHC, and Mites) and provides comprehensive detection results including pest classification, severity assessment, confidence scores, and location mapping. This integrated architecture ensures accurate pest identification while maintaining processing efficiency for real-time agricultural decision-making.

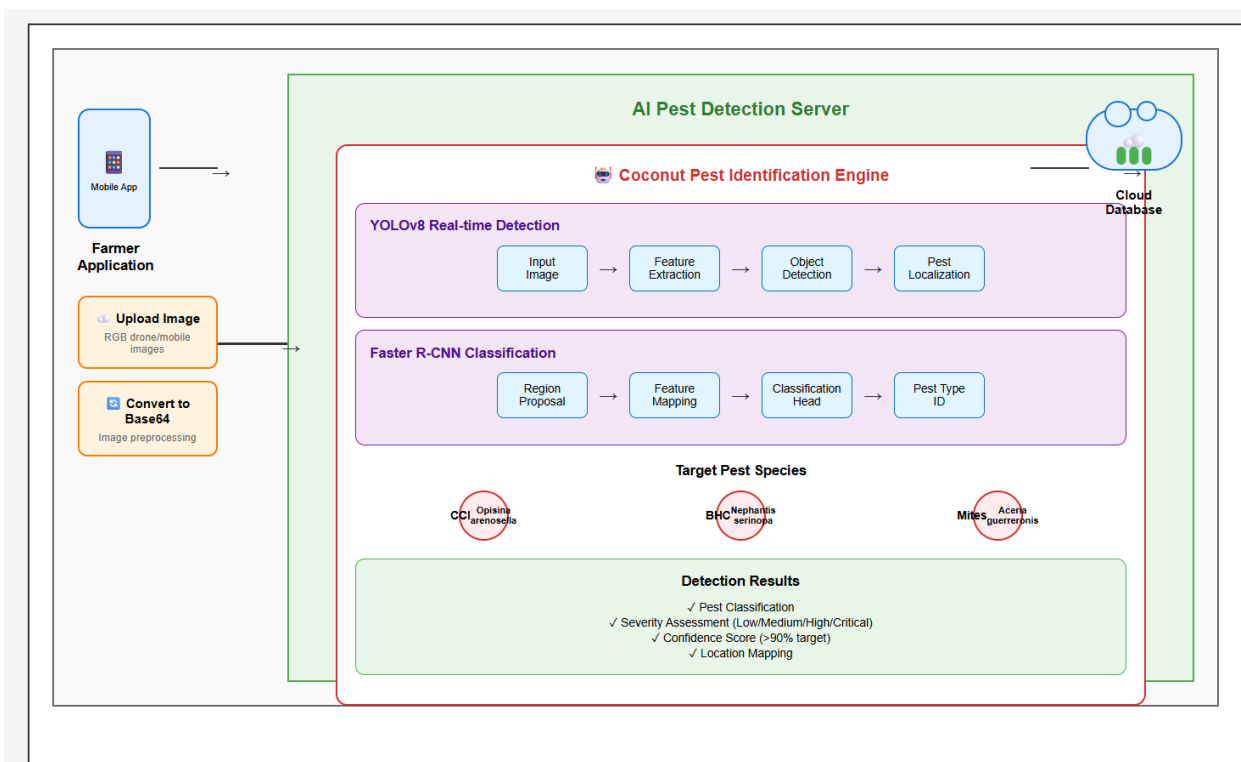


Figure 3: Identification of Coconut Pest Detection System Architecture

Figure 3 illustrates the overall high-level diagram of the proposed AI-powered pest detection component, which identifies pest infestations in coconut trees using advanced deep learning techniques. Initially the stakeholder will visit the mobile application. The stakeholder is provided with a choice to upload pre-existing images into the system or capture new imagery using a cross-platform mobile application that has been developed using React Native. The image that has been captured is subjected to preprocessing and subsequently uploaded to the cloud server using base64 string conversion. The converted image is then sent to the AI detection server, which uses dual

machine learning architectures - YOLOv8 for real-time object detection and Faster R-CNN for detailed pest classification - to determine the coconut pest infestation.

There are three subcomponents of the proposed component, as shown in Figure 3 First the system would identify whether there is a pest infestation present in the coconut tree imagery. Afterward, the system would identify the type of pest among the three target species (CCI, BHC, and coconut mites), and finally, the system would identify the severity level of the infestation caused by the detected pest. After identifying the pest type and severity, it will recommend the appropriate treatment measures and alert relevant stakeholders including farmers, extension officers, and nearby plantation owners.

Summary of technologies, techniques, architectures, and algorithms used for the classification of coconut pest infestations shown in the table below.

Technologies	Python, TensorFlow, PyTorch, OpenCV, Flask Server, React Native, MongoDB, AWS/Google Cloud
Techniques	Deep Learning, Computer Vision, Image Processing, Data Augmentation, Transfer Learning, Edge Computing
Algorithms	YOLOv8, Faster R-CNN, Non-Maximum Suppression (NMS), Anchor Box Generation
Architectures	ResNet-50 (Faster R-CNN backbone), YOLOv8 CSPDarknet, Feature Pyramid Network (FPN)

Table 3: Technologies, techniques, architectures, and algorithms used.

3.1.1 AI Development Solution

The AI Development Life Cycle (AIDLC) is the systematic process that machine learning teams use to design, develop, and deploy high-quality AI-powered solutions. This approach minimizes risks by incorporating iterative training, validation, and testing phases which enable the development team to meet accuracy expectations and performance requirements. In conventional AI methodology, developers need to complete extensive data collection and model training phases before deployment, making it difficult to adapt to changing requirements. To overcome this limitation, an agile AI development approach is adopted where the development team can iteratively improve model performance through continuous training and validation cycles. This

methodology is particularly suitable for computer vision applications like pest detection, where model accuracy can be incrementally improved through data augmentation and transfer learning techniques.

According to Figure 4, the AI Development Life Cycle for the pest detection system consists of six sequential phases that guide the systematic development of the AI-powered solution. The cycle begins with Data Collection & Preprocessing (Phase 1) where coconut pest images are gathered and prepared, followed by Model Architecture Design (Phase 2) where YOLOv8 and Faster R-CNN architectures are configured. Phase 3 involves Training & Validation of deep learning models using the collected datasets, while Phase 4 focuses on Integration & Deployment of the trained models into the mobile application. Phase 5 handles system Integration & Deployment across cloud infrastructure, and Phase 6 ensures continuous Performance Monitoring to maintain the target 90% detection accuracy. This iterative lifecycle approach enables continuous improvement of the pest detection system through ongoing validation and refinement.



Figure 4: AI Development Life Cycle for Pest Detection System

- **Requirement Gathering**

- **Data Gathering**

For data gathering, our team conducted systematic visits to multiple coconut plantations across key districts including Kurunegala, Gampaha, and Puttalam to collect comprehensive pest occurrence data. The farmers currently record pest data using manual observation methods and basic logbooks, which we collected as baseline information for system validation. Furthermore, we had to collect visual data manually by conducting drone flights over infested and healthy coconut trees to capture high-resolution RGB images. Additionally, we collaborated with the Coconut Research Institute of Sri Lanka (CRISL) and local Coconut Development Officers (CDOs) to gather expert knowledge on pest identification patterns, treatment protocols, and seasonal occurrence data.

- **Conducting a Survey**

We conducted a comprehensive survey by providing structured questionnaires to 242 coconut farmers across five districts to get a proper understanding of their current knowledge about pest identification, reporting mechanisms, and treatment practices. The survey revealed critical gaps in farmer awareness, with 72.7% of respondents indicating they do not know proper channels for reporting pest incidents to relevant authorities.

- **Feasibility Study (Planning)**

- **Economic Feasibility**

The proposed AI-powered pest detection component should operate efficiently without having computational errors or detection failures. The system will be more reliable and cost-effective compared to traditional manual inspection methods, reducing the need for frequent field visits by agricultural extension officers. By enabling early detection and targeted treatment recommendations, the system aims to reduce pesticide usage by 50% and crop losses by 40%, providing substantial economic benefits to coconut farming communities.

- **Scheduled Feasibility**

The proposed AI detection component has a direct impact on the overall project timeline, and it should be completed within the 8-month development period allocated. The component development follows a structured approach with data collection and annotation (2 months), model training and validation (3 months), system integration and testing (2 months), and field deployment (1 month). This component must achieve the target 90% detection accuracy within the specified timeline to ensure project completion by the deadline.

- **Technical Feasibility**

To complete this AI-powered solution, the development team requires specialized skills, resources and expertise in computer vision, deep learning architectures, drone operations, mobile application development, and cloud-based system deployment. Technical feasibility encompasses the required knowledge in Python programming, TensorFlow/PyTorch frameworks, YOLOv8 and Faster R-CNN architectures, image processing techniques, transfer learning methodologies, and mobile platform integration using React Native.

- **Design (System and Software Design Documents)**

After gathering requirements and feasibility study, we are moving on to the design phase. In this phase, comprehensive software documentation related to the AI system design is created, including detailed model architecture specifications, database schema design for pest occurrence data storage, mobile application user interface wireframes, cloud infrastructure deployment diagrams, and system integration protocols between drone image capture, AI processing server, and mobile application interfaces.

3.1.2 Use Case Diagram

According to Figure 5, the use case diagram illustrates the functional interactions between two primary actors and the pest detection system. The Coconut Farmer can register and login to the system, capture coconut tree images directly through the mobile camera, or upload existing images for analysis. Once images are processed, the system automatically assesses infestation severity levels and identifies specific pest types among the three target species (CCI, BHC, and Mites), then generates appropriate treatment recommendations based on the detection results. The System Administrator manages backend operations including sending alert notifications to relevant stakeholders and generating comprehensive pest reports for agricultural planning. The diagram shows include and extend relationships where user validation, error handling, and data processing occur seamlessly to provide farmers with immediate pest identification and treatment guidance while enabling administrators to monitor system performance and coordinate response activities.

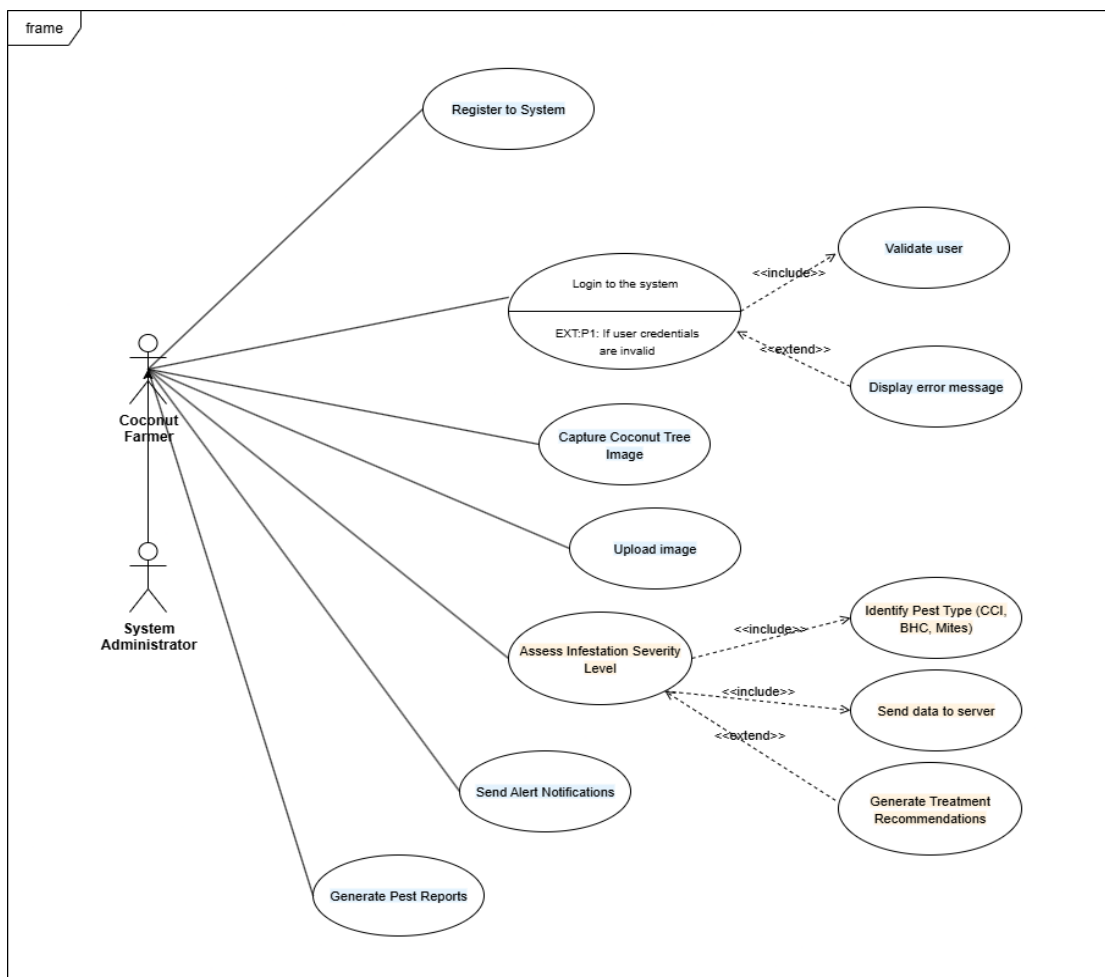


Figure 5: Use case diagram of Insect Infestation Identification component

3.1.3 Sequence Diagram

According to Figure 6, the sequence diagram demonstrates the step-by-step interaction flow for the pest identification process within the AI-powered detection system. The process begins when a stakeholder (farmer) logs into the system through the LoginUI and LoginController components, then proceeds to upload coconut tree images via the MainUI interface. The uploaded images are sent to the ImageController for preprocessing, followed by analysis through the AI server which processes the images using trained deep learning models. The system then displays the identified pest type through the PestTypeController and subsequently determines the infestation severity level via the SeverityController. Finally, the system analyzes and generates appropriate remedial measures based on the detected pest type and severity, providing farmers with comprehensive treatment recommendations. This sequential workflow ensures systematic processing from image upload to actionable pest management guidance, maintaining data flow integrity throughout the detection process.

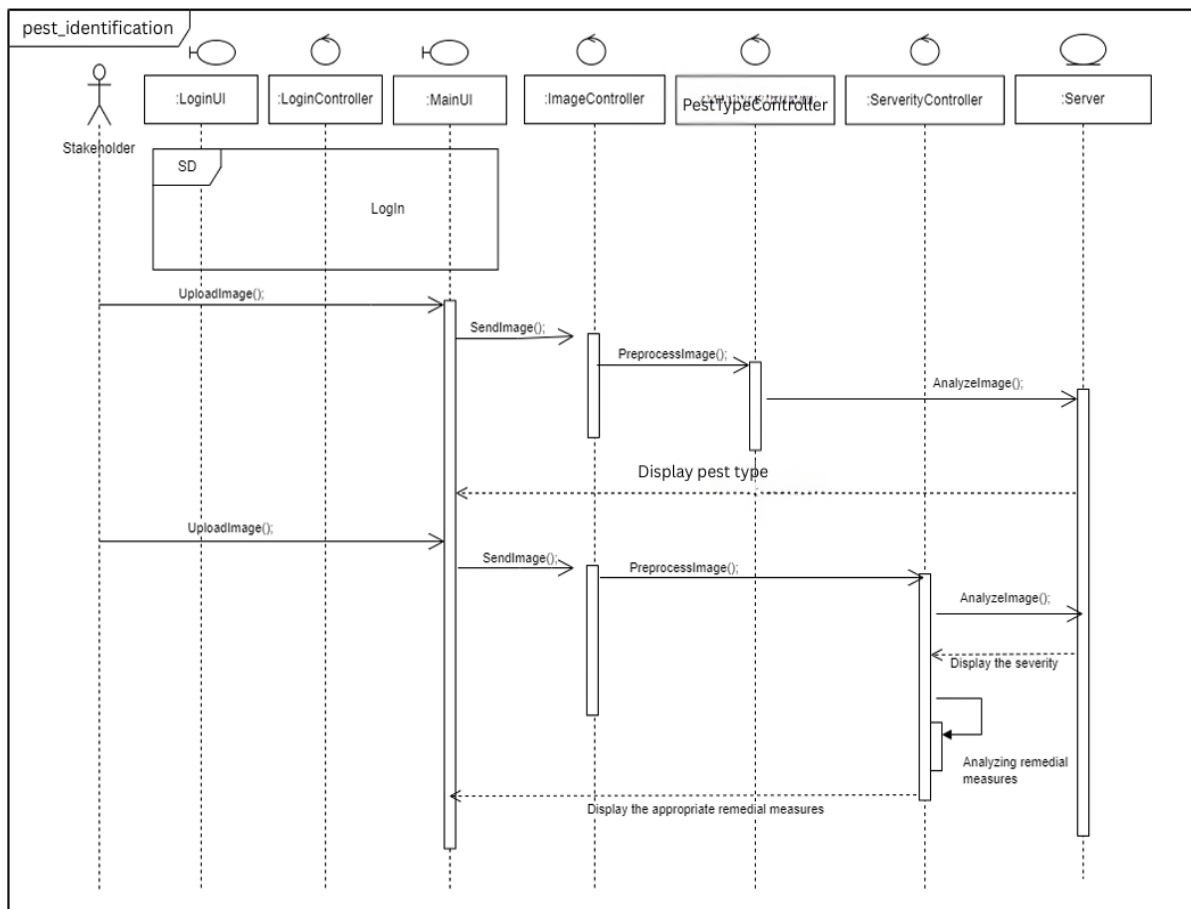


Figure 6: Sequence diagram of insect infestation identification component

3.1.4 Implementation and Development Sections

- **Implementation (Development)**

Below are the sub functionalities that will be implemented in this implementation phase.

- **Implementation of mobile application using React Native.** Cross-platform mobile application development enabling farmers to capture and upload coconut tree images with intuitive user interface design suitable for users with varying technological literacy levels.
 - **Implementation of AI pest detection engine using YOLOv8 architecture.** Real-time object detection model specifically trained on Sri Lankan coconut pest datasets to identify and localize pest infestations in drone-captured RGB imagery with processing speeds suitable for mobile deployment.
 - **Implementation of pest classification system using Faster R-CNN model.** Detailed classification component capable of distinguishing between three target pest species (CCI, BHC, and coconut mites) with confidence scoring and bounding box localization for accurate pest type identification.
 - **Implementation of severity assessment algorithm based on trained deep learning models.** An automated infestation severity evaluation system that analyzes pest density, affected leaf area, and damage patterns to classify severity levels as Low, Medium, High, or Critical for appropriate treatment recommendation.
 - **Implementation of cloud-based data processing and storage infrastructure.** Scalable cloud architecture for handling image uploads, AI model inference, result storage, and real-time synchronization between mobile applications and backend services.
- **Testing**
 - **Verifying that the AI detection system achieves the target 90% accuracy as expected.**
 - **Testing for model performance, processing speed, and integration compatibility.**

The testing of the AI-powered pest detection system will be conducted in various stages, including unit testing of individual model components, integration testing between mobile application and AI services, field testing under real-world coconut plantation conditions, and user acceptance testing with coconut farmers, with the latter being carried out by the target user community to ensure practical usability and effectiveness.

3.1.5 Commercialization

This research is proposed to develop a comprehensive AI-powered mobile application for coconut pest management in Sri Lankan agricultural communities. The target audience for the proposed system includes coconut farmers, agricultural cooperatives, plantation managers, drone service

providers, and agricultural technology companies. Our system consists of two deployment versions: a basic free version and a premium subscription version.

The features such as basic pest detection, severity assessment, and standard treatment recommendations will be provided in the free version. In the premium version, we will offer advanced analytics including historical trend analysis, predictive pest outbreak modeling, integration with drone service providers for automated plantation monitoring, detailed reporting dashboards, and priority customer support with expert consultation services.

When considering the marketplace, there is minimal competition for coconut-specific AI pest detection in Sri Lanka, requiring moderate technical knowledge for optimal usage, suitable for farmers aged 25-60, and minimal prior experience with AI-powered agricultural tools required. Our development team consists of technically qualified individuals with relevant academic backgrounds in computer science and artificial intelligence, with members gaining practical experience in agricultural technology applications.

3.1.6 Future Scope

For future scope, the application will be extended to integrate with IoT sensor networks and automated drone systems that could monitor coconut plantations continuously without human intervention. Additional enhancements may include integration with weather data APIs for predictive modeling, expansion to detect other coconut diseases beyond the current three pest types, multilingual support for regional languages, and potential integration with precision agriculture equipment for automated treatment application based on AI recommendations.

4 PROJECT REQUIREMENTS

4.1 Functional Requirements

1. The system should capture or upload coconut tree images using the smartphone or drone-captured imagery.
2. The system should identify the different pest types among three target species: Coconut Caterpillar Infestation (CCI), Black-Headed Caterpillar (BHC), and coconut mites.
3. The system should assess the severity level of pest infestation categorized as Low, Medium, High, or Critical based on AI analysis.
4. The system should recommend appropriate treatment measures including pesticide type, dosage, and application methods based on the detected pest type and severity level.

5. The system should provide real-time processing and deliver detection results with confidence scores exceeding 90% accuracy.
6. The system should store detection history and enable farmers to track pest occurrence patterns over time.

4.2 Non-functional Requirements

1. **User-friendliness** – Final goal is to create a mobile application that can be used on both Android and iOS platforms. The application should be visually appealing, farmer-friendly with intuitive navigation, and provide comprehensive pest detection features to users with varying technological literacy levels. Additionally, it should be designed to work seamlessly on both platforms, ensuring consistent experience for all coconut farmers.
2. **Reliability** – It is essential to protect all farmer data and pest detection information while ensuring users feel secure when utilizing the AI-powered detection system. Moreover, the system must maintain consistent accuracy in pest identification without encountering detection errors or system malfunctions that could lead to incorrect treatment recommendations.
3. **Performance** – The system should provide rapid and accurate pest detection results within 30 seconds of image upload while operating with optimal efficiency to meet farmer expectations during critical pest management decisions. The AI models should maintain processing speed suitable for field conditions.
4. **Availability** – The system must be consistently available to farmers whenever they require pest detection services, accessible through mobile connectivity across rural coconut plantation areas without limitations during peak agricultural seasons.
5. **Scalability** – The cloud infrastructure should support multiple concurrent users during pest outbreak periods and accommodate growing numbers of coconut farmers adopting the technology.

4.3 System Requirements

Required software and hardware requirements for the proposed system:

Software Requirements:

1. **VS Code** - To implement the AI detection algorithms in Python.
2. **Python 3.8+** - To implement the backend processing and AI model integration.

3. **TensorFlow/PyTorch** - To run the YOLOv8 and Faster R-CNN deep learning models.
4. **Flask Server** - To deploy AI models and handle API requests.
5. **React Native** - To design a cross-platform mobile application for farmers.
6. **MongoDB** - To store pest detection data, user information, and historical records.
7. **AWS/Google Cloud Platform** - For cloud-based model hosting and data storage.

Hardware Requirements:

1. **High-performance GPU** - For AI model training and inference processing.
2. **Smartphones with camera** - Android/iOS devices for image capture by farmers.
3. **Drone platforms** - Optional for aerial imagery capture in larger plantations.

4.4 User Requirements

This AI-powered pest detection application will be developed for the following user types:

1. Coconut Farmers:

The primary users will utilize the application to identify pest infestations by uploading images of coconut tree foliage captured through smartphones or drone imagery. Farmers will be able to receive immediate pest type identification, severity assessment, and specific treatment recommendations. Through this process, farmers can take timely and appropriate pest management actions, reduce crop losses and optimize pesticide usage for sustainable coconut cultivation.

2. System Administrators:

Technical administrators will use the application backend to monitor system performance, manage user accounts, and oversee AI model accuracy. They will be able to track detection statistics, update model parameters, and ensure system reliability. Administrators can gather performance data and maintain system optimization to provide consistent service quality to farming communities.

5 GANTT CHART

According to Figure 7, the Gantt chart presents the comprehensive 8-month project timeline spanning from June 2025 to March 2026, organized into six sequential phases with some overlapping activities. Phase 1 (Project Planning) runs from June to July covering project setup, literature review, and system architecture design. Phase 2 (Data Collection) occurs from August

to October focusing on drone image collection and dataset annotation. Phase 3 (AI Model Development) extends from October to December involving YOLOv8 and R-CNN training along with predictive analytics development. Phase 4 (Application Development) spans November to February encompassing backend API development, mobile application creation, and web dashboard implementation. Phase 5 (Testing & Validation) runs from January to March conducting AI model testing and field validation with farmers. Finally, Phase 6 (Documentation) occurs from February to March covering research paper writing, final report preparation, and presentation development. This structured timeline ensures systematic project execution with adequate time allocation for each critical development phase while maintaining parallel activities to optimize the 8-month duration.

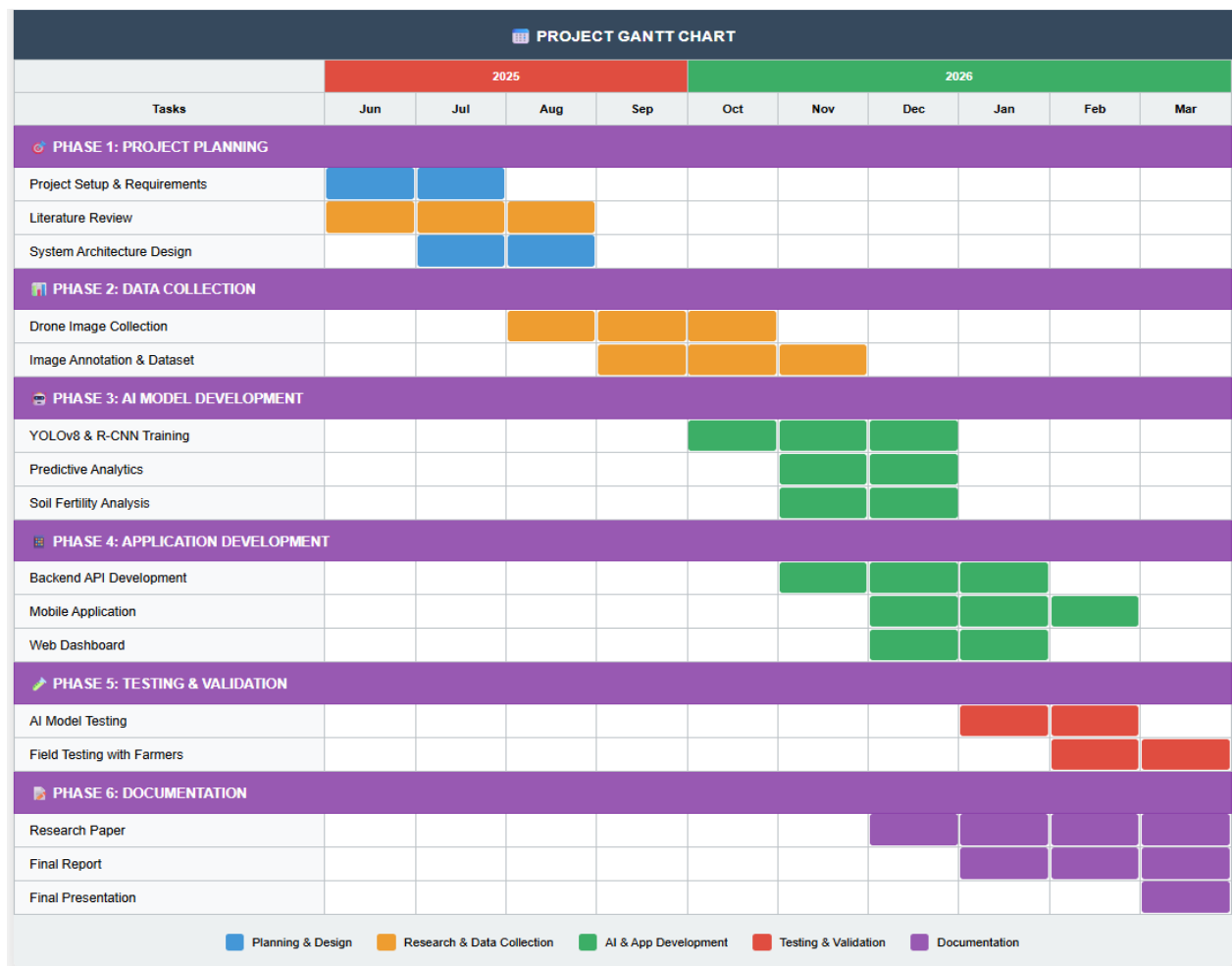


Figure 7 : Grant Chat

5.1 Work Breakdown Structure (WBS)

According to Figure 8, the Work Breakdown Structure (WBS) illustrates the hierarchical decomposition of the coconut pest identification project into six major phases and their respective sub-activities. The Initiation phase includes project charter creation, team formation, and stakeholder analysis. The Planning phase encompasses requirements analysis, literature review, feasibility study, and data collection planning. The Design phase covers system architecture development, AI model design, mobile app UI design, and database design. The Implementation phase involves deep learning model development, YOLOv8 and R-CNN training, mobile app development, and backend API development. The Testing phase includes AI model accuracy testing, field testing with farmers, system integration testing, and user acceptance testing. Finally, the Closeout phase covers research paper writing, final report preparation, final presentation, and system deployment. This structured breakdown ensures comprehensive project management by clearly defining deliverables and responsibilities for each phase of the AI-powered pest detection system development.

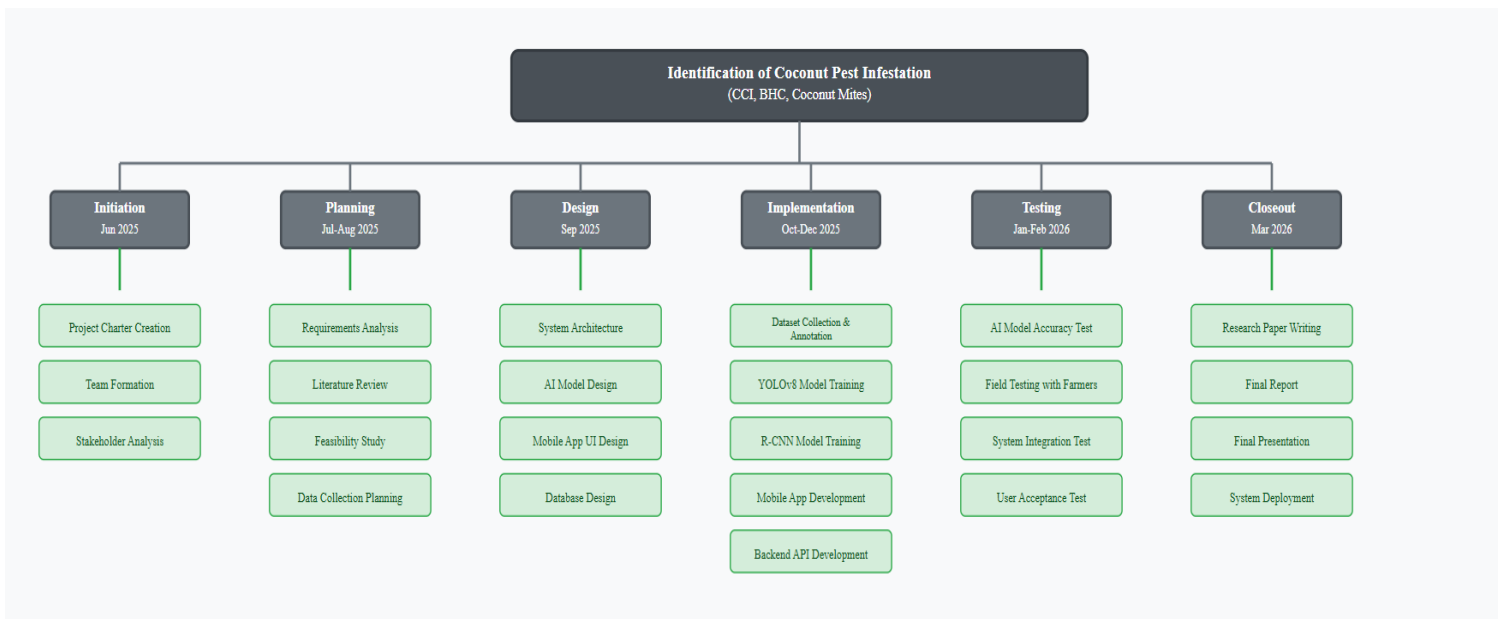


Figure 8: Work Breakdown Chart of identification of insect infestation

6 BUDGET AND BUGET JUSTIFICATION

According to Table 4, the project budget breakdown totals Rs. 135,000 distributed across six essential cost categories for the AI-powered coconut pest detection system development. Data collection represents the largest expense at Rs. 45,000 covering drone equipment and field data gathering activities, followed by cloud hosting and AI model training at Rs. 25,000 for computational resources required for YOLOv8 and Faster R-CNN model development. Research documentation and publication costs Rs. 18,000 for academic paper preparation and conference submissions, while mobile app development requires Rs. 15,000 for cross-platform React Native implementation. App Store hosting and maintenance accounts for Rs. 12,000 to ensure continuous system availability and traveling costs of Rs. 10,000 cover field visits to coconut plantations across multiple districts for data collection and farmer surveys. This budget allocation prioritizes core technical development activities while ensuring adequate resources for field validation and academic dissemination of research findings.

Requirement	Cost (Rs.)
Travelling cost for field visits and data collection	10,000.00
Data Collection (drone and equipment)	45,000.00
Cloud hosting and AI model training	25,000.00
Mobile app development and deployment	15,000.00
App Store hosting and maintenance	12,000.00
Research documentation and publication	18,000.00
Total Expenses	135,000.00

Table 4: Below table 4 shows the overall budget for the overall proposed system.

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